Manoj Poola

Data Scientist & ML Engineer

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SUMMARY

- **4+ years** of experience delivering data-driven and AI-powered solutions as a **Data Scientist** and **Machine Learning Engineer** across diverse industries, transforming complex datasets into actionable insights.
- Expert in **Python**, **SQL**, **NoSQL**, and advanced machine learning and data science frameworks, with a strong record of building end-to-end pipelines to streamline workflows and improve efficiency.
- Skilled in **Machine Learning**, **Deep Learning**, and **MLOps**, with expertise in TensorFlow, PyTorch, scikit-learn, Hugging Face Transformers, LangChain, and OpenAI APIs, focusing on robust model development, deployment, and hyperparameter tuning.
- Proficient in **Big Data Engineering** with Apache Spark, Hadoop, Airflow, and Amazon Redshift, enabling efficient processing of large-scale datasets.
- Hands-on experience with **Cloud Platforms** including **AWS** - S3, EC2, SageMaker, RedShift, Lambda, EMR, **GCP** - BigQuery, AI Platform, **Azure** - Data Lake, Machine Learning Studio, and **DevOps** - Docker, Kubernetes for deploying scalable ML applications in production environments.
- Expertise in **Statistical Analysis** and **Experimentation** using A/B Testing, Hypothesis Testing, Time-Series Analysis, PCA, and Bayesian Inference to support data-driven decision-making and model validation.
- Specialized in **NLP**, **Generative AI**, and **Computer Vision**, leveraging spaCy, NLTK, TF-IDF, Vector Databases, Transformers, and OpenCV to build multi-model models, automate workflows, and handle diverse data types.
- Skilled in **Data Visualization and Reporting**, creating dynamic dashboards and in-depth visualizations with **Tableau**, **Power BI**, Matplotlib, and Plotly for effective storytelling.
- **Collaborative Team Player**, experienced in partnering with cross-functional teams to implement innovative **AI-Driven Strategies**, optimize processes, and drive business success.

TECHNICAL SKILLS

Programming Languages:	Python, R, SQL, NoSQL, MATLAB, Java, C++, Spark, Hadoop, Git, Shell Scripting, JSON, XML, YAML
ML\DL Frameworks:	TensorFlow, PyTorch, scikit-learn, Keras, XGBoost, LightGBM, Hugging Face Transformers, Diffusers, LangChain, OpenAI GPT Models, Model Explainability (SHAP, LIME), Hyperparameter Optimization, Ensemble Methods
Data Engineering:	Apache Spark, Hadoop, Hive, Kafka, Airflow, Snowflake, Delta Lake, PostgreSQL, MySQL, MongoDB, Elasticsearch, ETL Pipelines, Amazon Redshift, Google BigQuery, Azure Data Factory
MLOps, Cloud & DevOps:	AWS (S3, EC2, SageMaker, Redshift, Lambda, EMR), GCP (BigQuery, AI Platform), Azure (Data Lake, Machine Learning Studio, Databricks), Docker, Terraform, GitLab CI/CD, MLflow
NLP and Computer Vision:	spaCy, NLTK, Gensim, Hugging Face Transformers, OpenAI APIs, TF-IDF, Vector Databases Word2Vec, BERT, GPT Models, YOLO, Vision Transformers (ViT), OpenCV, Object Detection, Image Segmentation
Data Visualization and Statistics:	Tableau, Power BI, Matplotlib, Seaborn, Plotly, QuickSight, Excel, Looker, D3.js, ggplot2, Grafana, Prophet

WORK EXPERIENCE

Texas A&M Health Science Center, College Station, TX, USA — **Jul 2024 – Present**
Data Scientist and Machine Learning Engineer

- Developed an automated pipeline to transfer brain imaging data to **AWS S3** storage using **Watchdog** for real-time file monitoring and triggering secure uploads via **AWS CLI** and **Boto3**, ensuring efficient data synchronization.
- Integrated bilinear averaging Algorithms and advanced image preprocessing with **Python** and **OpenCV**, including Gaussian Filtering, morphological transformations, and adaptive thresholding, achieving a 52% reduction in size.
- Combined **Lambda functions** for event-based triggers, **EC2** instances for batch transformations, and S3 lifecycle policies to lower monthly cloud storage bills from about $1,200 to under $600, driving cost-efficient data retention.

- Streamlined large-scale CSV datasets (over 4 million rows each) using **Pandas** and **Numpy** by reducing CPU load from 80% to 45% through vectorized operations, leading to quicker analysis cycles and smoother workloads.
- Deployed **Spark** on **Hadoop**-backed **Elastic MapReduce** to process ~32 TB of sensor logs in under 1 hour (down from 9+ hours), supporting real-time research updates for high-volume spatiotemporal data**.**
- Performed extensive data cleaning and automated Outlier Detection, Denoising and Normalization bringing error rates in seizure classification model down from ~12% to under 3%, improving overall dataset reliability and consistency.
- Implemented clustering algorithms such as **DBSCAN** for anomaly detection and **Gaussian Mixture Models** for **Clustering**, identifying time evolving behaviors in spatiotemporal datasets.
- Utilized **MLflow** for versioning across 20+ experiments and leveraged **SHAP** and **LIME** to interpret feature importance, enabling reproducibility and explainability in classification and clustering workflows.
- Created interactive visualizations using **Matplotlib** and **Plotly** to uncover patterns, support hypothesis generation, and drive actionable insights, while collaborating with researchers to enhance project outcomes.

UI Health and UICCE, Chicago, IL, USA **Jan 2023 – Jun 2024**
Data Scientist

- Designed and implemented automated **ETL** pipelines using **Python**, **SQL**, and **Apache Airflow** to unify and process over 150,000 patient records daily, reducing search response times from around 45 seconds to under 20 seconds.
- Developed a patient feedback capture system leveraging **NLP** techniques such as **Text Tokenization**, stop-word removal, **TF-IDF Vectorization**, and **Topic Modeling** to classify up to 2,000 pieces of feedback each week.
- Employed **spaCy** and **NLTK** to consolidate patient comments for over 60 physicians monthly. The automation replaced manual report generation, cutting a multi-day process to just a few hours with consistent action plans.
- Deployed **Cosine Similarity** and **Jaccard Index** on thousands of feedback records, flagging redundant or conflicting data resulting in reliable datasets and helping maintain a high standard of data hygiene with less manual intervention.
- Designed machine learning models, using **Random Forest** and **Support Vector Machines (SVM),** for predictive analytics through **Hyperparameter Tuning** and **Feature Selection**, boosted model **F1-score** from 0.70 to 0.88.
- Trained **Predictive Models** using **Scikit-learn** and **XGBoost** to identify readmission risks across multiple hospitals that reduced avoidable readmissions by an estimated 10% over a six-month pilot, streamlining resource allocation.
- Developed **Python**-based automation tools using **Selenium** for repetitive data analysis tasks tied to administrative operations, saving roughly 15 hours of clinical staff work time per week.
- Conducted advanced statistical analysis, including **Regression Models** and **Hypothesis Testing**, to validate core findings in real-world patient trials and contributed evidence-based insights.

Eamvey Technologies, Guntur, AP, India **Aug 2020 – Aug 2022**
Data Scientist

- Ingested approximately 15 million daily ad and customer records from Meta Ads using **API** into **Amazon S3**, accelerating cross-platform analytics and cutting data retrieval latency from ~10 minutes to under 3 minutes
- Orchestrated large scale multi-gigabyte advertising data using **Python** and **SQL** on **Amazon Redshift**, producing weekly insight reports that drove an additional $60K in annual revenue from optimized ad targeting.
- Orchestrated up to 10 simultaneous ad experiments weekly with **A/B Testing** framework deployed on **AWS SageMaker**, reducing average Customer Acquisition Cost (CAC) from $12 to $9 across multiple campaigns.
- Examined ~100 million time series based monthly impressions in **Amazon RDS** to refine daily ad scheduling, elevating overall click-through rates (CTR) from 3.2% to nearly 4%, a notable lift in engagement
- Built **Python** script to attribute conversions across 5+ advertising channels, deployed via **AWS Lambda** reducing channel-measurement discrepancies from ~15% to under 6%, enabling more accurate budget allocation.
- Optimized ad campaigns by combining Meta Ads data with demographics and behavioral insights from over 3 million consolidated records, pinpointing high-value audiences and boasting return on ad spend (RoAS) from $3.40 to $6.30.
- Built automated reporting pipelines using **AWS Lambda** and **Amazon QuickSight** to visualize campaign KPIs in real-time, eliminating manual data collation and saving analysts up to 10 hours per week.
- Collaborated with five cross-functional marketing squads to streamline campaign planning, reducing launch cycles from 14 days to under one week and enabling faster experimentation for sustained brand growth.

EDUCATION

University of Illinois at Chicago
Master of Science in Computer Science

SRM University
Bachelor of Technology in Computer Science and Engineering

CERTIFICATIONS AND ACHIEVEMENTS

AWS Certified AI Practitioner ongoing | Microsoft Technology Associate | Google Advanced Data Professional | Machine Learning – Stanford University

Guest lectures on AI and Data Science at leading universities | Transformed bootstrapped startup profitable in 4 months | Mentored over 600 students